

TASK-1-ZAHEER SK (23MID0195)

```
In [2]: def gender_features(word):  
        return {'last_letter':word[-1]}  
  
gender_features('ZAHEER')
```

```
Out[2]: {'last_letter': 'R'}
```

```
In [3]: from nltk.corpus import names  
names.words()  
print(len(names.words()))  
  
labeled_names = [(name,'male') for name in names.words('male.txt')] + [(name, 'female') for name in names.words('female.txt')]  
  
import random  
random.shuffle(labeled_names)  
  
featuresets = [(gender_features(n), gender) for (n,gender) in labeled_names]  
#Try Printing featurests  
  
train_set, test_set = featuresets[5000:], featuresets[:2000]  
  
import nltk  
classifier = nltk.NaiveBayesClassifier.train(train_set)  
  
classifier.classify(gender_features('David'))  
#classifier.classify(gender_features('Yann Lecun'))  
  
#print(nltk.classify.accuracy(classifier,test_set))
```

7944

```
Out[3]: 'male'
```

TASK-2

```
In [4]: from sklearn.feature_extraction.text import CountVectorizer  
vect = CountVectorizer(binary=True) # Explore other parameters too!!
```

```

corpus = ["Tesseract is good optical character recognition engine ", "optical character recognition is significant "]

vect.fit(corpus) #Learns the unique words (Vocab) in the corpus and gives indexing to it

#vect.vocabulary_ #stores the vocab #vocabulary_ built-in:-)
vocab = vect.vocabulary_

for key in sorted(vocab.keys()):
    print("{}:{}".format(key, vocab[key]))

print(vect.transform(["This is a good optical illusion"]).toarray()) #only after vect.fit only we can do transform

#Finding Similarity Between the Documents
from sklearn.metrics.pairwise import cosine_similarity
similarity = cosine_similarity(vect.transform(["Google Cloud Vision is a character recognition engine"]).toarray(), vect.trans
print(similarity)

```

```

character:0
engine:1
good:2
is:3
optical:4
recognition:5
significant:6
tesseract:7
[[0 0 1 1 1 0 0 0]]
[[0.89442719]]

```

TASK-3

```

In [5]: from urllib import request
url="http://www.gutenberg.org/files/2554/2554-0.txt"
response=request.urlopen(url)
raw=response.read().decode('utf-8')
print(type(raw))
print(len(raw))
raw[:75]

```

```

<class 'str'>
1135213

```

```
Out[5]: '*** START OF THE PROJECT GUTENBERG EBOOK 2554 ***\n\n\n\n\nCRIME AND PUNISHMENT\n'
```